

Product Carbon Footprint Study Report

Product Name: furniture, wooden

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: furniture, wooden

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg furniture, wooden

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg furniture, wooden.

2. System Boundary

furniture production, wooden, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of furniture, wooden produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **2.256 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

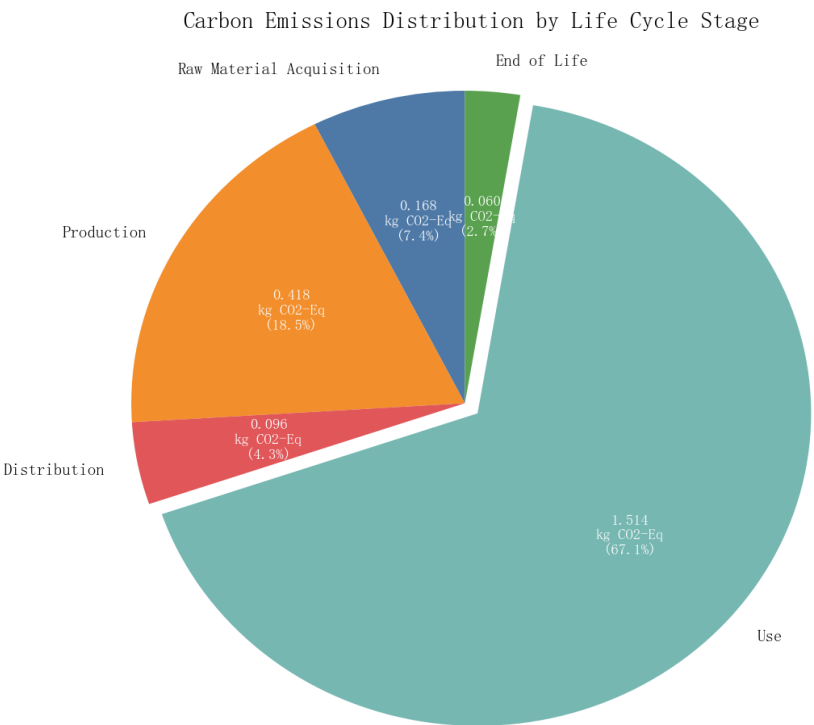


Table 1: Carbon Emissions by Life Cycle Stage

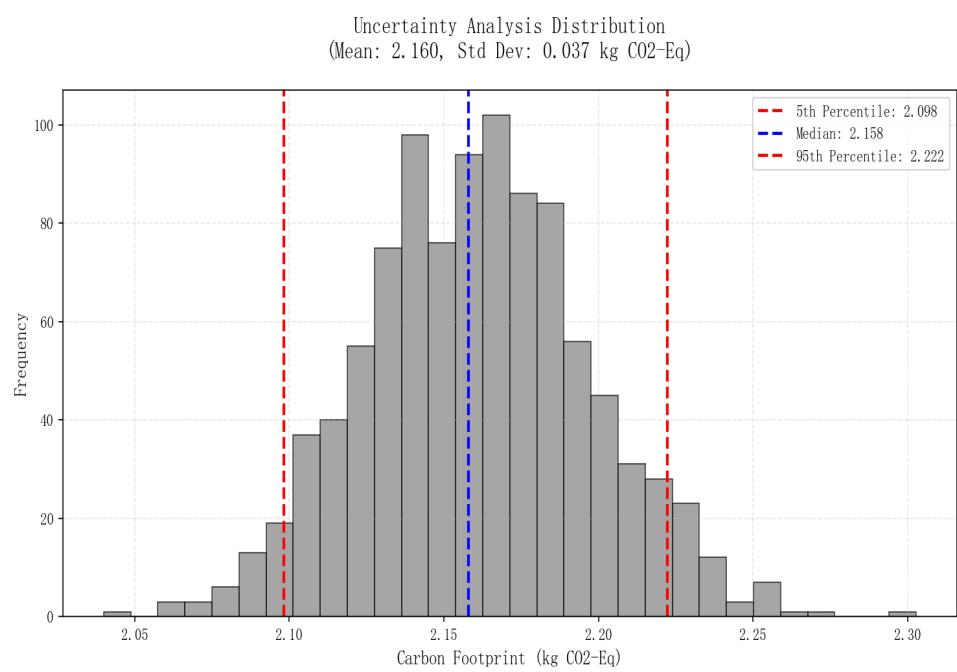
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.168	7.43
Production	0.418	18.53
Distribution	0.096	4.26

Use	1.514	67.11
End of Life	0.060	2.66
Total	2.256	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	2.160
Standard Deviation	0.037
Coefficient of Variation (CV)	1.71%
Median	2.158
5th Percentile	2.098
95th Percentile	2.222

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the wooden furniture product is calculated at 2.256 kg CO₂-Eq. This value represents a relatively low environmental impact per unit of furniture, typical for small to medium-sized wooden items, but context-dependent on product weight and lifespan. The lifecycle stage contributions reveal a highly skewed distribution, with the use phase dominating at 67.1% (1.514 kg CO₂-Eq). This indicates that the majority of emissions are associated with the product's operational lifetime, likely from energy consumption during cleaning, maintenance, or indoor climate control (e.g., heating/cooling due to furniture placement). The production stage contributes 18.5% (0.418 kg CO₂-Eq), reflecting manufacturing processes such as cutting, assembly, and finishing. Raw material extraction accounts for 7.4% (0.168 kg CO₂-Eq), suggesting efficient sourcing or low-impact forestry practices. Distribution (4.3%) and end-of-life (2.7%) have minimal contributions, implying short transport distances and low-emission disposal pathways. The major emission source is clearly the use phase, which warrants targeted investigation into user behavior and product durability.

2. Statement of Assumptions and Limitations

Key assumptions in this LCA study include a standardized use phase scenario, likely based on average consumer behavior over a typical product lifespan (e.g., 10–15 years). This assumes consistent maintenance activities (e.g., cleaning with energy-intensive appliances) and no significant energy rebound effects. The data sources for raw material and production stages are assumed to be secondary industry averages, with limited site-specific data, potentially underestimating variability in forestry and manufacturing practices. Limitations include the absence of detailed process contributions (all top processes listed as “Unknown”), indicating incomplete or aggregated input data that prevents granular hotspot identification. The system boundary is restricted to cradle-to-grave coverage, but excludes upstream capital goods (e.g., machinery) and downstream user behavior beyond standard assumptions. Additionally, biogenic carbon storage in wood is not explicitly modeled, which could offset some emissions if sustainably sourced. The uncertainty analysis shows a low coefficient of variation (1.71%) and a narrow 90% confidence interval (2.098–2.222 kg CO₂-Eq), suggesting high precision in the model output, but this precision may be misleading if underlying data gaps are not addressed.

3. Suggestions for Improvement

- ****Target the use phase:**** Implement product design changes to reduce user-phase emissions, such as integrating low-maintenance surfaces (e.g., easy-to-clean coatings) to minimize energy-intensive cleaning, or providing clear user guidelines for energy-efficient care (e.g., avoid vacuuming with high-wattage appliances).
- ****Optimize production processes:**** Invest in renewable energy sources for manufacturing facilities (e.g., solar or biomass) to cut the 18.5% production-stage emissions, and adopt waste-reduction techniques like closed-loop sawdust recycling.
- ****Enhance raw material sourcing:**** Prioritize certified sustainable forestry (e.g., FSC or PEFC) to lower raw material impacts, and consider using reclaimed or recycled wood to further reduce the 7.4% contribution.

- ****Improve data quality:**** Resolve the “Unknown” top process contributions by collecting detailed, supplier-specific data for all lifecycle stages, particularly for production and use-phase energy models, to enable precise hotspot analysis.
- ****Extend product lifespan:**** Promote repair, refurbishment, and modular design to prolong the use phase without increasing emissions, thereby reducing per-year carbon footprint and enhancing circularity.